Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course : B.Tech
Code : 105403

Subject: Operating System

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct answer of the following (any seven question only): **[2 x 7 = 14]**

(a) A_____ process is moved to the ready state when its time quantum expires.
(i) Blocked (ii) New
(iii) Running (iv) Suspended

(b) Which of the following is also called job scheduler?
(i) DMA controller (ii) Short-term scheduler
(iii) CPU scheduler (iv) Long-term scheduler

(c) Compaction is used to solve the problem of _____
(i) External fragmentation (ii) Internal fragmentation
(iii) both (i) and (ii) (iv) None of these

(d) Which process can be affected by other processes executing in the system?
(i) Independent process (ii) Co-operative process
(iii) Dependent Process (iv) child process

(e) What is the drawback of banker's algorithm?
(i) in advance processes rarely know how much resource they will need
(ii) the number of processes changes as time progresses
(iii) resource once available can disappear
(iv) all of the mentioned

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(f) An I/O port typically consists of four registers- status, _____, control, and _____registers.
(i) system in, system out (ii) flow in, flow out
(iii) data in, data out (iv) input, output

(g) The address loaded into the memory address register of the memory is referred to as_____.
(i) Physical address (ii) IP address
(iii) Port address (iv) Logical address

(h) Consider a disk queue with requests for I/O to blocks on cylinders.
98 183 37 122 14 124 65 67
Considering FCFS (first cum first served) scheduling, the total number of head movements is, if the disk head is initially at 53 is?
(i) 600 (ii) 630
(iii)620 (iv) 640

(i) The Trap mode bit changes from 0 to 1 when ____mode switches to ____mode
(i) Kernel, User (ii) User, User
(iii) Kernel, Kernel (iv) User, Kernel

(j) Waiting Time is calculated as _____
(i) Turnaround Time - Burst Time (ii) Turnaround Time – Arrival Time
(iii) Completion Time – Arrival Time (iv) Completion Time Burst Time

Q.2 (a) Define Thread? List some of the benefits of multithreading Demonstrate the three methods to implement Threads **[7]**

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(b) Assume the following workload in a system: [7]

Process	Arrival Time	Burst Time
P0	1	3
P1	0	6
P2	3	2
P3	4	7
P4	2	8
P5	6	5

Illustrating the execution of these job using Round Robin Scheduling algorithm calculate when given Time Quantum=3 :

(i) Average Turn Around Time
(ii) Average Waiting Time

Q.3 (a) Differentiate between: [9]

(i) User level Thread and Kernel level Thread
(ii) Preemptive scheduling and Non-preemptive scheduling
(iii) Contiguous and Indexed allocation method

(b) How is a process different from a program? With an example explain why Process synchronization is necessary in multiprocessing operating system? [5]

Q.4 (a) Describe the techniques for recovery from deadlock? Explain briefly resource allocation graph with examples. [7]

(b) Consider a disk queue with requests for I/O to blocks on cylinders. [7]
98 183 37 122 14 124 65 67
Considering SSTF (shortest seek time first) scheduling, the total number of head movements is, if the disk head is initially at 53 is?

Q.5 (a) What is the difference between fixed partition and variable partition also explains the need of compaction. [7]

(b) What is virtual memory? How is it implemented? Explain TLB in virtual memory with neat diagram. [7]

Q.6 (a) What is Reader-Writer problem? Explain how Semaphore is used to resolve the problem of process synchronization. [7]

(b) What is Belady's Anomaly? Why LRU and Optimal Page Replacement algorithm doesn't suffer from this problem. [7]

Q.7 Compare and contrast Deadlock Prevention and Deadlock Avoidance. Can a system be in a state that is neither deadlock nor safe? If so, give an example. If not, prove that all states are either deadlock of safe [14]

Q.8 Consider the following page reference string: [14]
1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6
Compare the page faults occur for the following replacement algorithms using four frames? Assume, initially all the frames are empty.

(i) LRU replacement
(ii) FIFO Replacement

Q.9 Write short notes on ***any two*** of the following:- [7 x 2 = 14]

(a) Dispatcher and Scheduler
(b) Starvation and Aging
(c) Paging and Segmentation
(d) Semaphore and Mutex

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